

Argument Principle.

Let f be a non-constant function analytic at z_0 with $f(z_0) = 0$.

Then, for some $r > 0$, f have only zero as z_0 in $B_r(z_0)$. Thus,

$$f(z) = (z - z_0)^k \cdot F(z) \text{ for some } k \in \mathbb{N},$$

and analytic function F at z_0 with $F(z_0) \neq 0$. Now,

$$f'(z) = k(z - z_0)^{k-1} F(z) + (z - z_0)^k \cdot F'(z),$$

this implies
$$\frac{f'(z)}{f(z)} = \frac{k}{z - z_0} + \frac{F'(z)}{F(z)}.$$

Since $F(z_0) \neq 0$, f'/f has simple pole at z_0 , with residue k . So,

$$\int_C \frac{f'(z)}{f(z)} dz = 2\pi i \cdot k.$$

Now, let f has a pole of order n at z_0 , then

$$f(z) = \frac{F(z)}{(z - z_0)^n}$$

where F is analytic at z_0 and $F(z_0) \neq 0$. Then,

$$f'(z) = -n \cdot \frac{F(z)}{(z - z_0)^{n+1}} + \frac{F'(z)}{(z - z_0)^n},$$

so therefore
$$\frac{f'(z)}{f(z)} = \frac{(z - z_0)^n}{F(z)} \cdot \left(-n \cdot \frac{F(z)}{(z - z_0)^{n+1}} + \frac{F'(z)}{(z - z_0)^n} \right) = \frac{-n}{z - z_0} + \frac{F'(z)}{F(z)}$$

so,
$$\int_C \frac{f'(z)}{f(z)} dz = 2\pi i \cdot (-n).$$

Obtain the two results, we obtain the following:

Thm. Let f be analytic inside and on a simple closed contour C except for finite poles in C , and $f(z) \neq 0$ on C . Then, counted with multiplicity, numbers of zeros N_f and poles P_f ,

$$\frac{1}{2\pi i} \cdot \int_C \frac{f'(z)}{f(z)} dz = N_f - P_f.$$

Example.

Let $f(z) = \frac{(z+1)(z+i)^5(z-i)^2}{(z^2-2z+2)^4(z+i)^8(z-5i)^8}$ and $C: |z|=2$.

Then, $z=-1, i, i$ are zeros of f inside C ,

and $z^2-2z+2=(z-1)^2+1$, so $1 \pm i$ and $-i$ are poles of order of 8 and 8.

Therefore, $\int_{|z|=2} \frac{f'(z)}{f(z)} dz = 2\pi i(3-16) = -26\pi i$.

Thm. Rouché Theorem.

Let f and g be analytic inside and on a simple closed contour C , with
 $|g(z)| < |f(z)|$ for all $z \in C$.

Then, $f+g$ and f have the same number of zeros inside C .

Example.

Let $p(z) = z^{10} - 6z^9 - 3z + 1$. Put $f(z) = -6z^9 + 1$ and $g(z) = z^{10} - 3z$.

Then, $|f(z)| = |-6z^9 + 1| \geq 6 - 1 = 5 \geq 1 + 3 \geq |z^{10} - 3z| = |g(z)|$ on $|z|=1$.

So, $p = f+g$ and f has number of roots in $|z|=1$, nine roots.

Example.

Let $p(z) = z^5 + 6z^3 + 2z + 10$. Put $f(z) = z^5 + 6z^3 + 2z$, $g(z) = 10$. Then, on $|z|=1$,

$|f(z)| = 9 < 10 = |g(z)|$, so $f+g$ has no root in $|z|=1$, same as g .

And, on $|z|=1$, $|p(z)| \geq 10 - 1 - 6 - 2 \geq 1$, so $p(z) \neq 0$ on $|z|=1$.

And, let $f(z) = z^5$ and $g(z) = 6z^3 + 2z + 10$. Then, on $|z|=3$,

$$3^5 \geq 6 \cdot 3^3 + 2 \cdot 3 + 10,$$

so all five zeros of p lie in $1 < |z| < 3$.